The Galois group of the 27 lines and the Brauer-Manin obstruction on cubic surfaces

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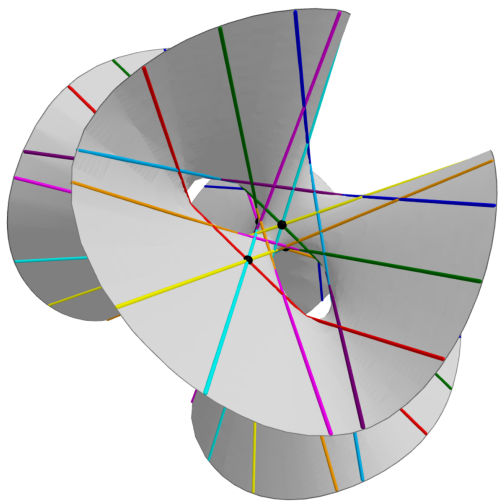
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A (smooth) **cubic surface** over k is a smooth projective hypersurface in $\mathbb{P}^3_k$ of degree 3.

Fun fact

Any cubic surface contains exactly 27 lines over the separable closure k_s. (And $\mathrm{Gal}(k_s/k)$ permutes these lines!)



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How do we check if the Hasse principle fails for a particular surface?

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$$X(\mathbb{A}_k)^{\mathrm{Br}} := \{(P_v)_v \in X(\mathbb{A}_k) : \sum_v \mathrm{inv}_v\, A(P_v) = 0 \ \forall A \in \mathrm{Br}(X)\},$$

where $\mathrm{Br}\, X := H^2_{\text{ét}}(X, \mathbb{G}_m)$.

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We say that there is a **Brauer-Manin obstruction** to the Hasse principle on X if we have $X(\mathbb{A}_k) \neq \varnothing$ but $X(\mathbb{A}_k)^{\mathrm{Br}} = \varnothing$.

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Conjecture (Colliot-Thélène, 1980)

For cubic surfaces[1], the Brauer-Manin obstruction is the only obstruction to the Hasse principle.

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Theorem (Swinnerton-Dyer, 1993)

Let X be a cubic surface over a global field k.
Suppose the action of $\mathrm{Gal}(k_s/k)$ *on the 27 lines of* X_{k_s} *is maximal*[2]*,*

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In particular, this implies that there is no Brauer-Manin obstruction on X! (Recall $\mathrm{Br}\, X/\mathrm{Br}\, k \simeq H^1(k, \mathrm{Pic}\, X_{k_s})$.)

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Main Theorem

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$\Downarrow$ Swinnerton-Dyer's Thm + Hilbert Irreducibility Thm

Corollary

Let k be a number field or $\mathbb{F}_q(t)$. Then 100% of cubic surfaces over k, ordered by height, have no Brauer-Manin obstruction.

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Remarks on the Main Theorem

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Remark

The characteristic 0 case of the theorem is classical and proven in various places, including by Jordan (1840) and Todd (1903). This was extended to large enough positive characteristics via formal arguments (e.g., in Banwait, Fité and Loughran, 2019). Another proof by Achter (2013) works for characteristics not equal to 2 or 3.

Our proof is different from all of these and uniform for all characteristics.

Crucially, we use downward induction on the degree of the del Pezzo surface.

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- Let $\mathscr{H}_d$ be the Hilbert scheme of del Pezzo surfaces of degree d with the anticanonical embedding into $\mathbb{P}^d$ (or a weighted projective space if $d \leq 2$).
- Let $\pi : \mathscr{Y}_d \to \mathscr{H}_d$ be the finite étale cover whose fiber above each point $X \in \mathscr{H}_d$ corresponds to the n_d exceptional curves on X.

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- Then the Galois group of the exceptional curves on the generic del Pezzo surface of degree d is equal to the image of the action of the fundamental group $\pi_1^{\text{ét}}(\mathscr{H}_d, x)$ on the fibers of π.

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- The case $d = 7$ is simple, because there are exactly two exceptional curves and it suffices to construct an example of a dP7 with nontrivial Galois action.

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- The induction proceeds by showing that the monodromy group in degree $d + 1$ is always a subgroup of the monodromy group in degree d. This, together with the transitivity, implies the result.